IL-NeRF: Incremental Learning for Neural Radiance Fields with Camera Pose Alignment

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Abstract

NeRF is a promising approach for generating photorealistic images and representing complex scenes. However, when processing data sequentially, it can suffer from catastrophic forgetting, where previous data is easily forgotten after training with new data. Existing incremental learning methods using knowledge distillation assume that continuous data chunks contain both 2D images and corresponding camera pose parameters, pre-estimated from the complete dataset. This poses a paradox as the necessary camera pose must be estimated from the entire dataset, even though the data arrives sequentially and future chunks are inaccessible. In contrast, we focus on a practical scenario where pre-estimated camera poses are unknown. We propose IL-NeRF, a novel framework for incremental NeRF training, to address this challenge. IL-NeRF's key idea lies in selecting a set of past camera poses as references to initialize and align the camera poses of incoming image data. This is followed by a joint optimization of camera poses and replay-based NeRF distillation. Our experiments on real-world indoor and outdoor scenes show that IL-NeRF handles incremental NeRF training and outperforms the baselines by up to 54.04% *in rendering quality.*

1. Introduction

Novel 3D representations, i.e., Neural Radiance Fields (NeRF) [32] and Gaussian Splatting [19], have recently shown great promise in producing photorealistic images from sparse image sets. To achieve this level of performance, they typically require access to all training data at once to estimate the camera pose based on the entire dataset [9, 19, 32]. However, in practical applications such as automotive and remote sensing, data is acquired sequentially, necessitating an immediately available updated 3D scene representation. Moreover, scenarios arise where a user acquires scans of a scene to train view synthesis models, only to find that the training yielded less effective results than ex-

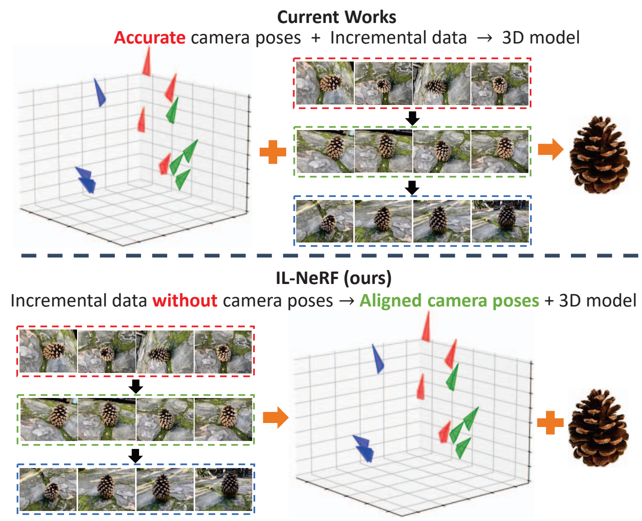


Figure 1. Current works require accurate camera poses pre-estimated from the whole image data. Our IL-NeRF can incrementally learn the 3D scene without these pre-estimated camera poses. The results of IL-NeRF consist of both aligned camera poses and NeRF model.

pected. Consequently, the user rescans the scene with new data to enhance the fidelity of the images rendered by these view synthesis models. In these instances, the scene representation must be trained in an incremental training environment, where the models can only access a limited number of views at each training stage, while still undertaking the task of reconstructing the entire scene.

To the best of our knowledge, existing works [6, 9, 15, 39] focusing on novel 3D representations in incremental learning scenarios have predominantly centered around NeRF. To have a fair comparison with these approaches, this paper also delves into incremental learning based on NeRF. However, our incremental learning method is designed to be adaptable to Gaussian Splatting as well, as we will demonstrate in subsequent experiments. In the context of incremental NeRF training, existing works [6, 9, 15, 39] generally operate under the assumption that data is segmented into multiple sequential chunks, with access limited to the current chunk while previously processed data is discarded. This assumption to incremental learning presents a notable challenge for NeRF, which requires updating its knowledge without erasing prior learned information, a phenomenon known as catastrophic forgetting [60]. To address this challenge, prior works [6, 9, 15, 39] have investigated the implementation of incremental learning strategies for NeRF training, incorporating a knowledge distillation technique [60] to mitigate catastrophic forgetting. Specifically, before training the NeRF model with the current data chunk, pseudo-RGB values for the scene are generated using the previously trained NeRF model. These RGB values are subsequently utilized to train the NeRF model with the current data chunk, and the process is iteratively repeated, with the prior NeRF model acting as the supervisory teacher for subsequent data chunks. This enables the NeRF model to learn from new data while retaining knowledge from previously discarded data.

Motivation. While existing works propose effective incremental learning methods through knowledge distillation, these approaches are on the assumption that continuous data chunks comprise not only 2D images *but also corresponding camera pose parameters, pre-estimated from the whole dataset*. This assumption poses a paradox as the required camera pose must be estimated from the entire dataset, yet the data is intended to arrive sequentially, relatively independently, with future data chunks remaining unknown and inaccessible while previous chunks are discarded. *In contrast, as shown in Figure 1, our work addresses a more practical scenario where pre-estimated camera poses from the whole dataset are unavailable for each training data chunk, necessitating the consideration of camera pose estimation and the alignment of its coordinate system.*

Challenge. Since the previous training data have been discarded, the incoming training data cannot simply be used directly for camera pose estimation because the isolated estimated camera pose will not be in the same coordinate system as the previous camera pose, which will lead to NeRF training misalignment and failure to render the 3D scene. Therefore, accurately estimating the camera poses of the sequential coming data within the same coordinate system in incremental NeRF training becomes a crucial issue that needs to be addressed.

Contribution. To deal with the above challenge, we present a novel framework for incremental NeRF training, named IL-NeRF, which can incrementally estimate the incoming data's camera poses and effectively tackle the issue of catastrophic forgetting. (1) We propose an incremental camera pose alignment module that selects a suitable set of camera poses from previous estimates. These chosen poses help render prior training images from NeRF and aid in the joint camera pose estimation process. They also act as a reference coordinate system, facilitating the alignment of newly arrived and previous camera poses. (2) To ensure the appropriate selection of camera poses from the previously estimated ones, we transform this selection task into a graph-based reward-collection optimization problem. We then introduce a practical greedy algorithm to effectively solve this optimization problem. (3) To align camera pose coordinates, we use selected camera poses as a reference to derive a transfer matrix, transforming the current camera poses into the previous coordinate system. (4) We utilize a joint optimization method for camera poses and replay-based NeRF distillation, mitigating catastrophic forgetting and refining the accuracy of the camera poses.

The experimental results on diverse datasets show that our proposed framework can improve PSNR, SSIM, and LPIPS by up to 54.04% compared to the original NeRF, significantly mitigating the negative impact of catastrophic forgetting in NeRF's incremental learning process. Moreover, our framework can effectively estimate and align the camera pose parameters in a consistent coordinate system.

2. Related Works

3D Representations. Various 3D representations have been proposed for different 3D tasks. NeRF [32] employs volume rendering to depict a continuous scene and achieve high-quality view synthesis. Several subsequent works have been introduced based on the success of NeRF, aiming to improve view synthesis efficiency and quality, including faster training and rendering [7, 12, 34], deformable or dynamic scene synthesis [37, 40, 41], editable view synthesis [35, 55], light changes [30, 33], surface enhancements [36, 51, 58], depth priors [10, 42, 54], etc. Recently, 3D Gaussian Splatting [19] has significantly enhanced rendering speed to real-time levels by representing scenes as 3D Gaussians. This technique finds application in various domains such as 3D or 4D reconstruction [27, 57], avatar modeling [21, 24], and Gaussian Splatting-based generation [8, 48], etc. However, most of these methods presume access to all training data and require pre-estimated camera pose parameters from the entire dataset. In this study, we tackle a more practical scenario where 3D Representations learn the scene incrementally with a sequential data stream, without pre-estimated camera pose parameters. Here, NeRF serves as a representative 3D representation for comparison with existing NeRF-based incremental learning methods. Nevertheless, our proposed incremental learning approach is adaptable to 3D Gaussian Splatting, as we will demonstrate in subsequent experiments.

Incremental NeRF Learning. Incremental learning, constrained by limited data availability during each training iteration, often causes catastrophic forgetting [11]. Methods to mitigate this issue include parameter isolation [16, 28, 29], replay [26, 43, 46], and regularization [1, 2, 56]. There is limited research on combining NeRF with incremental learning. Existing studies [6, 9, 15, 39] use knowledge distillation [60] to mitigate catastrophic forgetting by accessing past training data, specifically RGB values, from a pre-trained network. The retrieved data is merged with new incoming data to train NeRF. In [15], a regularization-based filter is used to select relevant information from randomly sampled camera views. In [9], a small network is trained to generate previously seen rays that are directed toward the scene. The authors in [39] employ the same replay-based method in [9] but they substitute NeRF with Instant-NGP [34] to expedite the training process. In [6], a prioritized replay buffer is introduced to keep the images and their camera poses with the lowest historical rendering qualities for continual learning. However, these methods require pre-estimated camera poses from the entire training data for each coming data chunk. In this work, we consider a more realistic scenario where pre-estimated camera poses are unavailable for each training data chunk, thus requiring incremental camera pose alignment.

NeRF with Pose Refinement. Pose refinement is widely used in NeRF training. iNeRF [59] refines camera poses for novel view images using a reconstructed NeRF model. NeRFmm [53] jointly optimizes both camera intrinsic and extrinsic parameters during NeRF training, and BARF [25] proposes a coarse-to-fine positional encoding strategy for camera poses and NeRF joint optimization. SC-NeRF [18] further considers camera distortion refinement and employs a geometric loss to regularize rays. NoPe-NeRF [5] integrates mono-depth maps with the joint optimization of camera parameters and NeRF, which can achieve a good rendering quality and pose estimation accuracy. In this paper, IL-NeRF uses the pose refinement in SC-NeRF [18]. Note that IL-NeRF is not limited to only the pose refinement in SC-NeRF [18]; any other well-designed pose refinement methods can also be transferred to IL-NeRF.

3. Incremental NeRF Training Preliminary

NeRF. NeRF takes 3D location $\mathbf{x}$ and view direction $\mathbf{r}_d$ as input and produces RGB color $\mathbf{c}$ and volume density σ as output. For each ray $\mathbf{r} = (\mathbf{r}_o, \mathbf{r}_d)$ emitted from the camera origin $\mathbf{r}_o$ in direction $\mathbf{r}_d$, NeRF samples M 3D points along the ray $\mathbf{x}_i = \mathbf{r}_o + z_i\mathbf{r}_d$, where z_i is the distance from a camera to the sample point $\mathbf{x}_i$. Thus the pixel color C can be integrated by the volumetric rendering as follows:

$$C(\mathbf{r}) = \sum_{i=1}^{M} \alpha_i (1-\delta_i)\mathbf{c}(\mathbf{x}_i) \tag{1}$$

where $\delta_i = \exp(-(z_i - z_{i-1})\sigma(\mathbf{x}_i)$ represents the transmittance of the ray segment between sample points $\mathbf{x}_{i-1}$ and $\mathbf{x}_i$, and $\alpha_i = \prod_{j=1}^{i-1}\delta_j$ is the ray attenuation from the origin $\mathbf{r}_o$ to the sample point $\mathbf{x}_i$. Since the whole pipeline is differentiable, NeRF can be trained by minimizing the photometric error between the rendering views and ground truth views.

$$\mathcal{L} = \sum_{\mathbf{r}\in R} \parallel C^*(\mathbf{r}) - C(\mathbf{r}) \parallel_2^2 \tag{2}$$

$$\Theta^* = \arg\min_{\Theta} \mathcal{L}(C \mid C^*, \mathcal{P}) \tag{3}$$

where R represents a group of rays from one or more camera views, which is obtained from the camera pose parameters $\mathcal{P}$. C^* is the ground truth of pixel color. Θ denotes the parameters of the network. For more details of NeRF, we refer the readers to [32].

Incremental NeRF Training. We consider a time-slotted system wherein each time slot $t \in \{0, \cdots, T\}$, a chunk of image data G^t consists of N number of images I^t, i.e., $G^t = \{I_n^t, n \in \{0, \cdots N-1\}\}$. Generally, only the latest image data chunk G^t is available while previous image data chunks $G^{0:t-1}$ are inaccessible. The objective of incremental NeRF is to minimize reconstruction loss across all provided chunks of image data in $\{G_0, \ldots, G_T\}$. Note that unlike the existing works [6, 9, 15, 39], in our work, each incoming data contains only the images, without corresponding camera poses. *Thus, in this paper, the main aim of incremental learning for NeRF is to enable the neural network to learn and adapt continually by ensuring that the camera poses from new image data are estimated and aligned in a consistent coordinate system while preventing catastrophic forgetting across all previously seen image data.*

4. IL-NeRF

In this section, we introduce our proposed framework, IL-NeRF, which prevents the catastrophic forgetting problem by replay-based knowledge distillation retrieved from NeRF itself (Section 4.1) and utilizes the incremental camera pose alignment to estimate and align the camera poses of incoming training image data within the same coordinates system as the previous camera poses (Section 4.2). The overall pipeline is illustrated in Figure 2.

4.1. Replay-Based NeRF Distillation

The problem of catastrophic forgetting occurs when a network, trained only with the currently available data at each time step, struggles to remember previously learned knowledge, resulting in low-quality image synthesis for previously seen views. To address this, we adopt a replay-based NeRF distillation strategy in the NeRF training. At each time slot t, we copy and freeze the parameters of the network as a teacher network before training the network on the incoming image data chunk t. Since the network has been trained on $t-1$ previous image data chunks, we use Θ_{t-1}^* to denote the frozen parameters of the teacher network. During each training iteration for the image data chunk t, we use the past camera poses $\mathcal{P}^p$ to obtain the pixel colors of the past training rays from the teacher network Θ_{t-1} as follows:

$$C^p = \mathcal{F}(\mathcal{P}^p, \Theta_{t-1}^*) \tag{4}$$

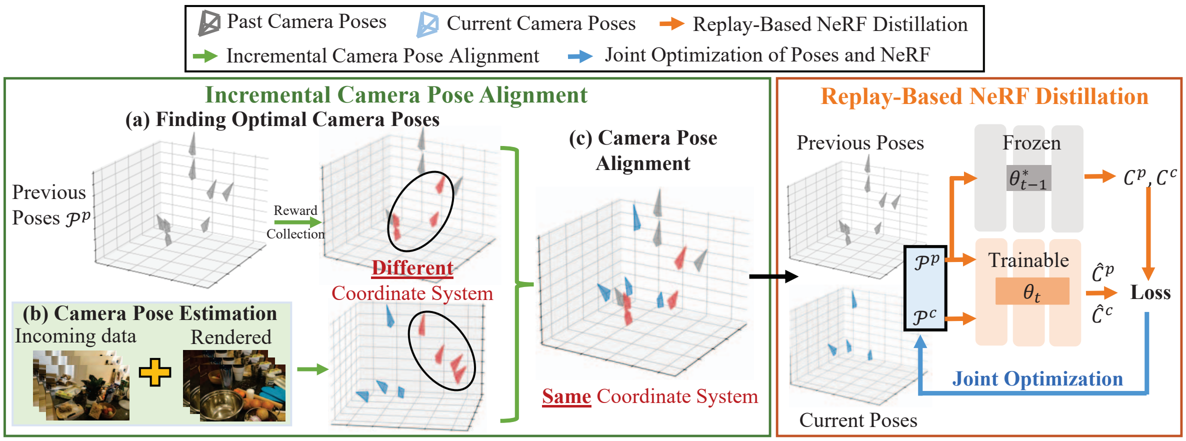


Figure 2. Overview of IL-NeRF pipeline. Firstly, the network Θ^*_{t-1} from the previous NeRF is frozen. Then, incremental camera pose alignment is employed to estimate the current camera poses $\mathcal{P}^c$ through (a) Finding optimal camera poses from the previous camera poses; (b) Estimating the camera poses for the incoming image data and the rendered images from the selected camera poses; (c) Aligning the current camera poses into the previous camera coordinate system. Finally, the network Θ_t, the current estimated poses $\mathcal{P}^c$, and previous poses $\mathcal{P}^p$ are jointly trained on both the current image data rays C^c and the distilled past rays C^p simultaneously.

By treating C^p as pseudo ground truth, we optimize Θ_t by learning new knowledge from the new incoming image data and old knowledge from the teacher network, thus mitigating the forgetting problem. The new loss function is defined as follows:

$$\mathcal{L} = \sum_{\mathbf{r}^c} \| \hat{C}^c - C^c \|_2^2 + \sum_{\mathbf{r}^p} \| \hat{C}^p - C^p \|_2^2 \qquad (5)$$

$$\Theta^*_t = \arg\min_{\Theta_t} \mathcal{L}(\hat{C}^c, \hat{C}^p \mid C^c, \mathcal{P}^c, \Theta^*_{t-1}, \mathcal{P}^p) \qquad (6)$$

where $\mathbf{r}^c$ and $\mathbf{r}^p$ are the training rays obtained from current camera poses $\mathcal{P}^c$ and past camera poses $\mathcal{P}^p$. $\hat{C}^c$ and $\hat{C}^p$ are the estimated colors by the network that is being trained, given the concatenated camera poses $[\mathcal{P}^c, \mathcal{P}^p]$:

$$[\hat{C}^c, \hat{C}^p] = \mathcal{F}([\mathcal{P}^c, \mathcal{P}^p], \Theta_t) \qquad (7)$$

The existing works [6, 9, 15, 39] assume that the camera poses are provided with each incoming image data and not saved on the device when the next round of training image data arrives. This requires additional effort to train a new auxiliary neural network to remember or filter the previously trained rays. However, this paper considers a different scenario. Specifically, the incoming image data chunk only includes the training images and not the camera poses, and hence the camera poses need to be obtained using camera calibration techniques and saved on the device. It's worth noting that only the previous camera poses are stored, not the previous training images. Furthermore, the camera pose requires only a small storage space. In the following section, we will discuss how to estimate and align the previous and new camera poses into the same coordinate system.

4.2. Incremental Camera Pose Alignment

Let $\mathcal{P}^p$ represent the previously aligned camera poses. which includes intrinsic camera matrix K and extrinsic camera matrices $\{\pi^p, \psi^p\}$, where π^p is the set of rotation matrices and ψ^p is the set of translations. When it comes to estimating camera poses from the incoming image data, it is not sufficient to assume that the problem can be resolved by independently estimating the camera poses of the incoming image data. This is because the outcomes derived from the isolated estimation of camera poses do not align with the coordinate system of the previous camera poses, leading to a significant issue of coordinate misalignment. To this end, we introduce an incremental camera pose alignment method leveraging data from a trained NeRF. We start by choosing D camera poses from prior instances based on low training losses and comprehensive coverage. These are then combined with incoming images to enhance camera estimation.

Finding Previous Optimal Camera Poses. Our approach is to formulate a reward-collection optimization problem on a graph. In this graph, the nodes represent camera positions (i.e., the translations in the camera poses) with each node assigned a reward corresponding to the negative value of the preceding training loss. The edges represent Euclidean distances between each camera pair's positions. The goal is to find a path that collects as much reward as possible, subject to constraints on the total number of visited nodes and camera view coverage. The objective optimization problem can be formulated as

$$\max \sum_{k=1}^{|\mathcal{P}^p|} x_k R_k; \quad s.t. \sum_{k=1}^{|\mathcal{P}^p|} x_k = D; \qquad (8)$$

$$S(K) \geq S_{th}, K = \{k | x_k = 1, k = 1, \dots, |\mathcal{P}^p|\}; \quad (9)$$

$$E(x_k) \leq 1, \forall k \in \{1, \dots, |\mathcal{P}^p|\}; \qquad (10)$$

where x_k is the binary decision variable: $x_k = 1$ if node k is visited otherwise $x_k = 0$. $S(K)$ is the shortest path that connects all the selected nodes. $E(x_k)$ is the number of incoming edges of each selected node. The first constraint (8) makes sure that only D previous camera poses are selected. The second constraint (9) means that the view coverage of the selected cameras is larger than a threshold, because a large field of view coverage of the selected cameras improves the accuracy of the camera pose estimation. The third constraint (10) guarantees that every node only has one incoming edge. In other words, every node is visited at most once. Consequently, this reward-collection optimization problem can be viewed as a hybrid of the Knapsack Problem and the Travelling Salesperson Problem, which is an NP-hard problem. To solve this problem, we propose a greedy algorithm that can reduce computation time by several orders of magnitude compared with the Brute-Force method. Let $e_{i,j} = e_{j,i}$ denote the edge between camera i and j. At camera node i, we define an approximation edge length of all the nodes connecting to the node i as $\hat{e}_{i,j} = R_j + \lambda(\frac{S_{th}}{D} - e_{i,j})$, where λ is a parameter for adjusting the units of R_i and $\frac{Sth}{D} - e_{i,j}$. This approximation edge is similar to the Lagrange multiplier [4] for handling the constraint (9). We introduce an auxiliary starting node into the graph, which connects to all nodes with the same edge length. The greedy algorithm begins at this auxiliary starting node and selects the unvisited node with the maximum approximation edge as the next starting node. This process is repeated until a total of D nodes have been selected. For a comprehensive understanding of the process, we offer a detailed description of the greedy algorithm, along with pseudocode, in the supplementary material.

Camera Pose Alignment. After solving the above reward-collection optimization problem, we select D camera poses with large view coverage from the previously aligned camera poses. The reason for selecting D camera poses instead of using all the camera poses is that camera poses with poorly rendered images will provide inaccurate features leading to large camera estimation errors.

These D camera poses are utilized to render the images from the NeRF model, which are integrated with the incoming image data as a group. The camera poses of this group can be estimated using camera calibration methods. Let π_d be the rotation matrix of the selected camera in time slot $t-1$ and $\tilde{\pi}_d$ be the corresponding rotation matrix of the selected camera in time slot t. Similarly, let ψ_d be the translation of the selected camera in time slot $t-1$ and $\tilde{\psi}_d$ be the corresponding translation of the selected camera in time slot t. We can then get the transfer matrices of the rotation matrix and translation from the coordinate system in time slot t to the coordinate system in time slot $t-1$ by:

$$\triangle\pi = \frac{1}{D}\sum_{d=1}^{D}\pi_d(\tilde{\pi}_d)^{-1}, \triangle\psi = \frac{1}{D}\sum_{d=1}^{D}(\psi_d - \triangle\pi\tilde{\psi}_d) \quad (11)$$

We can use transfer matrices to align the camera poses $\{\pi, \psi\}$ of the new images to the original camera poses by:

$$\tilde{\pi} = \triangle\pi\pi, \quad \tilde{\psi} = \triangle\pi\psi + \triangle\psi \quad (12)$$

Joint Optimization of Poses and NeRF. However, we have observed that the camera pose alignment still produces errors in the camera poses. To address this, we draw inspiration from previous works [18, 53] and introduce a joint optimization of poses and NeRF method during training of our proposed IL-NeRF. Rather than directly optimizing the initial camera pose $\tilde{\pi}$ and $\tilde{\psi}$, we employ a 6-dimensional vector $\Phi = [a, b]$ to define the trainable parameters of each camera pose, where $a \in \mathbb{R}^3$ represents the 3D rotation angles and $b \in \mathbb{R}^3$ denotes the increment of the translation. To ensure the orthogonality of the rotation matrix, Rodrigues' formula $\Omega(a)$ is used to generate the 3D rotation matrix. Final rotation and translation are:

$$\tilde{\pi} = \Omega(a)\tilde{\pi}, \quad \tilde{\psi} = \tilde{\psi} + b \quad (13)$$

By integrating the 6-dimensional vector Φ into the NeRF training pipeline, the camera parameters and scene representation can be jointly optimized during training. Here, we slightly abuse the notation to use Φ to represent the group of all cameras' trainable parameters. Mathematically, the pose refinement can be written as:

$$\Theta_t^*, \Phi_t^* = \arg\min_{\Theta_t, \Phi_t} \mathcal{L}(\hat{C}^c, \hat{C}^p \mid C^c, \mathcal{P}^c, \Theta_{t-1}^*, \mathcal{P}^p) \quad (14)$$

5. Experiment

Dataset. We use three different datasets to evaluate different aspects of our model, namely Mip-NeRF360 [3], NeRF-real360 [32] and Block-NeRF [47]. To simulate the incremental scenarios, we select a portion of the dataset where the previous images are not revisited. Following existing works [6, 9, 15, 39], all the training images of each scene are divided into four training chunks denoted as $\mathcal{G} = \{G^0, G^1, G^2, G^3\}$.

Baseline and Metrics. IL-NeRF is compared with the following baselines: **NeRF**: Original NeRF is incrementally trained with only current image data chunk with ground truth camera poses but without NeRF distillation, making it susceptible to catastrophic forgetting. **EWC** [20]: Similar to the NeRF, EWC incrementally trains the model with only current image data chunk with ground truth camera poses however it utilizes a widely-used regularization-based method, which penalizes the changes in parameters that are important for past training sets. EWC and original NeRF are the common baselines in all existing works [6, 9, 15, 39]. **CLNeRF** [6]: NeRF is incrementally trained with ground truth camera poses under replay-based NeRF distillation. *Note that CLNeRF requires the ground truth camera poses in each incoming image data chunk, which are estimated from all the training images.* **NeRF-SLAM** [44]: we follow the general implementation of NeRF-SLAM, which use the SLAM to align camera poses of the coming image data.

We evaluate IL-NeRF and baselines in three aspects, including Peak Signal-to-Noise Ratio (PSNR), Structural

Table 1. Performance comparison with the baselines on PSNR, SSIM, and LPIPS. IL-NeRF outperforms the original NeRF, EWC, NeRF-SLAM and achieves comparable results with CLNeRF. Note that CLNeRF, NeRF, and EWC require the ground truth pre-estimated camera poses from entire image data, but IL-NeRF estimates and aligns camera poses by the proposed incremental camera pose alignment module. **PE Pose implies whether the incremental data chunk carries the camera poses pre-estimated in advance through the entire dataset.**

Data		Method	PE Pose	PSNR ⇑ / SSIM ⇑ / LPIPS ⇓			
				G^0	G^1	G^2	G^3
Mip-NeRF360	Counter	CLNeRF	Yes	32.17 / 0.92 / 0.07	29.58 / 0.86 / 0.14	28.03 / 0.82 / 0.18	28.28 / 0.85 / 0.18
		NeRF	Yes	32.12 / 0.91 / 0.07	24.62 / 0.72 / 0.25	21.94 / 0.65 / 0.34	20.30 / 0.62 / 0.37
		EWC	Yes	32.12 / 0.91 / 0.07	23.83 / 0.72 / 0.25	22.56 / 0.65 / 0.33	21.11 / 0.61 / 0.36
		NeRF-SLAM	No	31.75 / 0.91 / 0.07	28.30 / 0.83 / 0.21	26.84 / 0.79 / 0.28	25.30 / 0.77 / 0.31
		IL-NeRF	No	32.13 / 0.91 / 0.07	29.63 / 0.87 / 0.12	28.56 / 0.85 / 0.15	27.82 / 0.83 / 0.17
	Kitchen	CLNeRF	Yes	31.05 / 0.91 / 0.07	29.72 / 0.88 / 0.13	29.33 / 0.85 / 0.15	29.18 / 0.84 / 0.14
		NeRF	Yes	31.17 / 0.91 / 0.08	27.01 / 0.75 / 0.25	21.42 / 0.70 / 0.31	23.69 / 0.75 / 0.24
		EWC	Yes	31.17 / 0.91 / 0.08	26.76 / 0.74 / 0.25	22.09 / 0.70 / 0.31	23.39 / 0.74 / 0.23
		NeRF-SLAM	No	30.87 / 0.90 / 0.09	29.63 / 0.85 / 0.20	27.65 / 0.81 / 0.24	27.71 / 0.82 / 0.20
		IL-NeRF	No	31.27 / 0.92 / 0.07	30.66 / 0.89 / 0.10	29.84 / 0.87 / 0.12	29.34 / 0.86 / 0.13
NeRF-real360	Pinecone	CLNeRF	Yes	26.88 / 0.89 / 0.12	24.23 / 0.79 / 0.16	24.03 / 0.73 / 0.19	23.18 / 0.74 / 0.21
		NeRF	Yes	26.22 / 0.84 / 0.16	22.90 / 0.64 / 0.24	21.15 / 0.58 / 0.33	18.94 / 0.49 / 0.41
		EWC	Yes	26.22 / 0.84 / 0.16	22.70 / 0.63 / 0.24	21.42 / 0.57 / 0.32	18.81 / 0.48 / 0.41
		NeRF-SLAM	No	25.63 / 0.81 / 0.18	24.09 / 0.73 / 0.22	23.01 / 0.68 / 0.29	21.79 / 0.65 / 0.34
		IL-NeRF	No	26.3 / 0.87 / 0.10	24.56 / 0.78 / 0.17	23.78 / 0.74 / 0.20	22.93 / 0.72 / 0.23
	Vasedeck	CLNeRF	Yes	29.27 / 0.86 / 0.07	27.93 / 0.85 / 0.12	26.03 / 0.74 / 0.16	26.18 / 0.74 / 0.18
		NeRF	Yes	29.03 / 0.85 / 0.07	23.99 / 0.70 / 0.26	22.73 / 0.69 / 0.24	21.57 / 0.64 / 0.31
		EWC	Yes	29.03 / 0.85 / 0.07	24.36 / 0.69 / 0.25	22.25 / 0.68 / 0.24	20.52 / 0.64 / 0.30
		NeRF-SLAM	No	27.98 / 0.79 / 0.11	26.41 / 0.77 / 0.21	25.10 / 0.72 / 0.21	24.62 / 0.71 / 0.26
		IL-NeRF	No	29.48 / 0.86 / 0.07	27.38 / 0.82 / 0.10	26.11 / 0.76 / 0.14	26.15 / 0.75 / 0.17

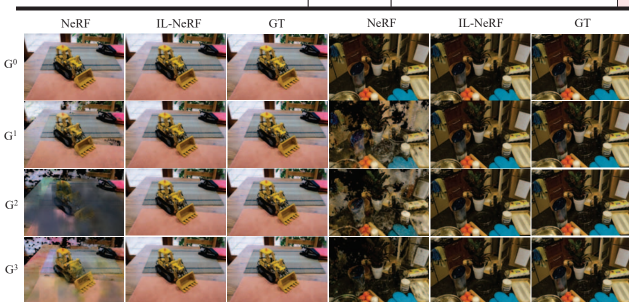


Figure 3. Qualitative comparison of the original NeRF and IL-NeRF on the rendering images in the first image data after each incremental training. GT means the ground truth of the training image. The original NeRF demonstrates severe catastrophic forgetting, leading to the loss of early-task scene information. In contrast, IL-NeRF can preserve the scene of interest throughout the training process.

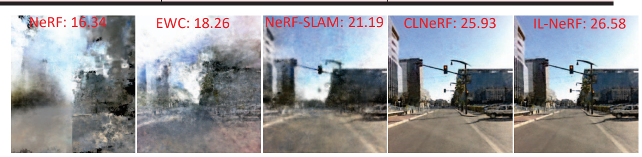


Figure 4. PSNR comparison on Block-NeRF dataset.

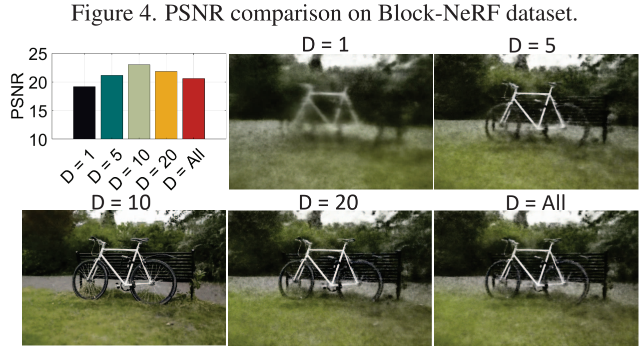


Figure 5. Influence of optimal pose count D on IL-NeRF.

Similarity Index Measure (SSIM) [52] and Learned Perceptual Image Patch Similarity (LPIPS) [61]. We use AlexNet [22] as the backbone of LPIPS. Note that as joint optimization of poses and NeRF is performed for all camera poses, including the previous and current poses, at each time step, the camera poses are continuously changing. This means it is not possible to fix the camera poses for test data, and all evaluation metrics compare the rendered images from the training camera poses with the ground truth.

5.1. Results

Table 1 shows the partial results obtained by IL-NeRF and baseline methods on the three datasets. Due to page limitations, *more comparisons are shown in the supplementary material*. From the results, we can see that IL-NeRF outperforms the original NeRF, EWC, NeRF-SLAM and achieves comparable results with CLNeRF. We explain the results in more detail next.

NeRF: In comparison to the original NeRF, IL-NeRF demonstrates substantial enhancements of between 10.87% to 36.55% in terms of PSNR, 15.18% to 46.90% in terms of SSIM, and 25.00% to 54.05% in terms of LPIPS across three datasets. In the initial image data chunk G^0, IL-NeRF outperforms the original NeRF slightly, as a result of joint optimization assistance provided by IL-NeRF. However, the performance of the original NeRF rapidly declines thereafter due to catastrophic forgetting.

EWC: EWC fails to reduce the adverse effects of catastrophic forgetting, and in some cases can be even worse than traditional NeRF. This is due to the lack of previous training images, the failure of EWC to recover previous scenes, and the introduction of a penalty mechanism that creates a disincentive to learn new images that have not been scanned.

CLNeRF: Comparing CLNeRF with IL-NeRF may not be entirely fair, given that CLNeRF benefits from having access to all training images to estimate camera poses for the image data chunks, while in our scenario, the camera poses are not provided in the task and must be derived through incremental camera pose alignment. Nonetheless, IL-NeRF performs comparably to CLNeRF, due to its incremental camera pose alignment module utilized during training.

NeRF-SLAM: While NeRF-SLAM uses SLAM to align camera poses in incoming image data to a common coordinate system, it lags behind our IL-NeRF in terms of performance. This difference stems from NeRF-SLAM's exclusive reliance on selected keyframes for replay-based training, resulting in overfitting to specific rays and compromising multiview consistency. Furthermore, NeRF-SLAM demands more memory storage space. For instance, NeRF-SLAM necessitates an additional average 251.3 MB of memory for storing keyframes and point clouds. In contrast, IL-NeRF only requires an additional average 37.7 KB of memory for storing previous camera poses.

Figure 3 provides additional insights by presenting a qualitative comparison of the performance of the original NeRF and IL-NeRF. Specifically, we demonstrate the rendering results on the first image data after each incremental training. It is evident that the original NeRF suffers from the catastrophic forgetting problem, resulting in images with significant distortions such as noise and blur, whereas IL-NeRF generates highly realistic images with quality comparable to the ground truth. This observation indicates that IL-NeRF is highly effective in mitigating the forgetting problem and estimating the camera poses. *More results and video comparison of IL-NeRF and all baselines are shown in the supplementary material*.

Large Scale Dataset. To further show the performance of IL-NeRF, we compare IL-NeRF with baselines on large scale dataset Block-NeRF [47]. As we can see in Figure 4, IL-NeRF also has a good performance on large-scale dataset Block-NeRF.

5.2. Ablation Study

Effect of Pose Selection. The first step of incremental camera pose alignment is to find previous D optimal camera poses as the references for estimating and aligning the camera poses of the incoming image data. To illustrate the influence of D on the IL-NERF performance, we set the value of D to 1, 5, 10, 20, and all, respectively. Figure 5 portrays the PSNR of IL-NeRF across varying values of D on the 'Bicycle' scene from the Mip-NeRF360 dataset. As depicted, when D is a small value, the lack of adequate reference images results in the estimated camera poses of the incoming image data deviating from the original camera pose coordinate system, thereby leading to considerably poor rendering. As the value of D increases, the estimated camera poses of the incoming image data become increasingly precise, and thus the PSNR increases. Note that an excessively large value of D introduces poorly rendered cameras, subsequently leading to a decrease in PSNR. To identify the optimal D camera poses, we address a reward-collection optimization problem on a graph. To demonstrate the effectiveness of our camera selection approach, we conduct a comparative analysis with two baselines: (1) Randomly selecting D camera poses as the reference, and (2) Myopically selecting D camera poses with the lowest training losses. Figure 6 shows the performance of IL-NeRF and two baselines on the 'Bicycle' scene. Our proposed method surpasses the other two approaches. This is primarily due to our method's ability to ensure the quality of rendered images used as references while providing a broader camera view coverage, thereby facilitating the more accurate camera pose estimation of incoming image data. *More results are shown in the supplementary material*.

Effect of Transfer Matrices. The transfer matrices are obtained by computing the corresponding rotation matrix and translation of the selected D camera poses in time slots $t-1$ and t. These matrices are then employed to align the camera pose of new images to the original camera pose coordinate system. To investigate the effectiveness of the transfer matrices, we compare IL-NeRF with IL-NeRF without considering the transfer matrices, denoted as 'IL-NeRF w/o TM' in Figure 7. The results reveal a significant decline in performance without the transfer matrices, achieving only $15.76dB$ in terms of PSNR. This decline can be attributed to separate camera pose estimation for two tasks resulting in camera poses in two independent coordinate systems, which could mislead the model during training, leading to decreased performance. *More results are shown in the supplementary material*.

Effect of Pose Refinement. Despite the transfer matrices' ability to align the camera poses to the original coordinate system, they may still contain noise and inaccuracies. To mitigate this issue, we use the coordinate-aligned camera poses as initial values and jointly optimize the camera poses and scene representation during NeRF training, a process known as pose refinement. We perform an ablation study to investigate the effectiveness of pose refinement by comparing IL-NeRF with and without it. The results in Figure 7 indicate that IL-NeRF with pose refinement outperforms IL-NeRF without it (i.e., IL-NeRF w/o PR). *More results are shown in the supplementary material.*

5.3. Camera Pose Alignment

Our goal is to incrementally train a NeRF model given only RGB images as input, without known camera poses. In other words, we need to find out the camera poses associated with each input image while training the NeRF model. We treat the COLMAP estimation from all training images as ground-truth (GT) camera poses and report the difference between our optimized camera poses and theirs on the training images. Figure 9 shows the camera pose trajectories of GT and IL-NeRF. As we can see, IL-NeRF recovers accurate camera poses with the help of camera coordinate alignment and pose refinement. *More results are shown in the supplementary material.*

Combining CLNeRF with Our Camera Pose Alignment. Table 2 shows the results of CLNeRF with provided camera poses (CLNeRF), CLNeRF with our camera pose alignment (CLNeRF-CPA). Comparison with CLNeRF and CLNeRF-CPA, our camera pose alignment is not only limited to IL-NeRF, but also can be combined with existing methods to solve more practical incremental learning scenarios.

Combining Gaussian Splatting with Our Camera Pose Alignment. Table 2 also compares Gaussian Splatting with provided camera poses (GS) with Gaussian Splatting with our camera pose alignment (GS-CPA). GS-CPA employs the same camera pose selection and camera pose alignment, but for pose refinement, we refer to the pose refinement method outlined in [13]. As we can see, our camera pose alignment technique is effective not only for NeRF but also for Gaussian Splatting in incremental learning scenarios.

Comparing with Existing Pose Optimization in NeRF. We use NoPe-NeRF [5] as an example and replace our pose camera alignment with NoPe-NeRF. We keep camera poses of previous data estimated by NoPe-NeRF, and when new incremental data arrives, we use the previous camera poses as the initial values, so that we can ensure that the camera poses of the new incremental data can be consistent with the coordinate system of the previous camera poses. Table 2 also shows PSNR of NoPe-NeRF and IL-NeRF. IL-NeRF outperforms NoPe-NeRF. This is because NoPe-NeRF needs the previous data rendered from NeRF for depth estimation. Some poor quality replay data brings poor depth estimation results, which affects the performance of NoPe-NeRF.

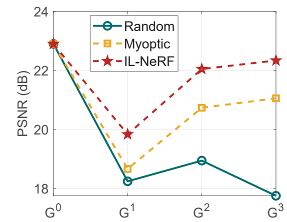


Figure 6. IL-NeRF outperforms random selection and myopic selection.

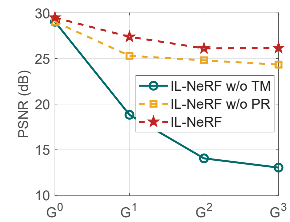
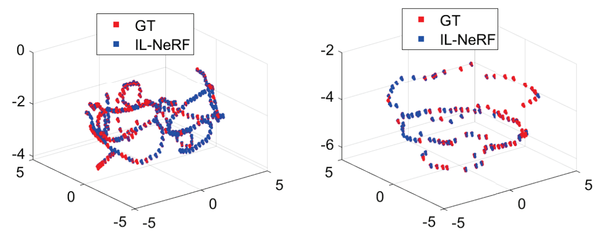


Figure 7. IL-NeRF outperforms IL-NeRF w/o TM and IL-NeRF w/o PR.

Figure 8. Comparison of IL-NeRF w/o Transfer Matrices (TM), IL-NeRF w/o Pose Refinement (PR) and IL-NeRF.



(a) Kitchen (b) Vasedesk

Figure 9. Camera pose estimation comparison. GT means the camera poses estimated by COLMAP from all the training images. IL-NeRF recovers accurate camera poses with the help of incremental camera pose alignment.

Table 2. Comparison of CLNeRF, CLNeRF-CPA, NoPe-NeRF, GS, GS-CPA and IL-NeRF on PSNR. The higher the better.

Method	Mip-NeRF360		NeRF-real360		Block-NeRF
	Counter	Kitchen	Pinecone	Vasedeck	
CLNeRF	**28.28**	29.18	**23.18**	**26.18**	25.93
CLNeRF-CPA	27.96	28.87	23.15	25.23	25.79
NoPe-NeRF	21.38	19.81	17.65	19.36	18.97
IL-NeRF	27.82	**29.34**	22.93	26.15	**26.58**
GS	29.15	31.64	24.88	27.73	26.91
GS-CPA	29.72	30.96	24.27	28.02	26.89

6. Conclusion

In this study, we introduce an incremental learning algorithm called IL-NeRF that tackles the problems of catastrophic forgetting and coordinate shifting in NeRF training in incremental learning settings. The IL-NeRF algorithm employs a replay-based NeRF distillation pipeline to retain past information and learn from new data independently. Furthermore, a method for aligning camera pose coordinates is introduced to identify camera poses associated with incoming tasks during the NeRF model training. Experimental results reveal that the IL-NeRF algorithm outperforms the original NeRF model in sequential data settings.

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References

[1] Rahaf Aljundi, Punarjay Chakravarty, and Tinne Tuytelaars. Expert gate: Lifelong learning with a network of experts. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 3366–3375, 2017. 2

[2] Rahaf Aljundi, Francesca Babiloni, Mohamed Elhoseiny, Marcus Rohrbach, and Tinne Tuytelaars. Memory aware synapses: Learning what (not) to forget. In *Proceedings of the European conference on computer vision (ECCV)*, pages 139–154, 2018. 2

[3] Jonathan T Barron, Ben Mildenhall, Dor Verbin, Pratul P Srinivasan, and Peter Hedman. Mip-nerf 360: Unbounded anti-aliased neural radiance fields. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5470–5479, 2022. 5

[4] Dimitri P Bertsekas. *Constrained optimization and Lagrange multiplier methods*. Academic press, 2014. 5, 2

[5] Wenjing Bian, Zirui Wang, Kejie Li, Jia-Wang Bian, and Victor Adrian Prisacariu. Nope-nerf: Optimising neural radiance field with no pose prior. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4160–4169, 2023. 3, 8, 4

[6] Zhipeng Cai and Matthias Müller. Clnerf: Continual learning meets nerf. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 23185–23194, 2023. 1, 2, 3, 4, 5

[7] Anpei Chen, Zexiang Xu, Andreas Geiger, Jingyi Yu, and Hao Su. Tensorf: Tensorial radiance fields. In *Computer Vision–ECCV 2022: 17th European Conference, Tel Aviv, Israel, October 23–27, 2022, Proceedings, Part XXXII*, pages 333–350. Springer, 2022. 2

[8] Zilong Chen, Feng Wang, and Huaping Liu. Text-to-3d using gaussian splatting. *arXiv preprint arXiv:2309.16585*, 2023. 2

[9] Jaeyoung Chung, Kanggeon Lee, Sungyong Baik, and Kyoung Mu Lee. Meil-nerf: Memory-efficient incremental learning of neural radiance fields. *arXiv preprint arXiv:2212.08328*, 2022. 1, 2, 3, 4, 5

[10] Kangle Deng, Andrew Liu, Jun-Yan Zhu, and Deva Ramanan. Depth-supervised nerf: Fewer views and faster training for free. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12882–12891, 2022. 2

[11] Robert M French. Catastrophic forgetting in connectionist networks. *Trends in cognitive sciences*, 3(4):128–135, 1999. 2

[12] Sara Fridovich-Keil, Alex Yu, Matthew Tancik, Qinhong Chen, Benjamin Recht, and Angjoo Kanazawa. Plenoxels: Radiance fields without neural networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5501–5510, 2022. 2

[13] Yang Fu, Sifei Liu, Amey Kulkarni, Jan Kautz, Alexei A. Efros, and Xiaolong Wang. Colmap-free 3d gaussian splatting. 2023. 8, 4

[14] Bruce L Golden, Larry Levy, and Rakesh Vohra. The orienteering problem. *Naval Research Logistics (NRL)*, 34(3): 307–318, 1987. 1

[15] Mengqi Guo, Chen Li, and Gim Hee Lee. Incremental learning for neural radiance field with uncertainty-filtered knowledge distillation. *arXiv preprint arXiv:2212.10950*, 2022. 1, 2, 3, 4, 5

[16] Ching-Yi Hung, Cheng-Hao Tu, Cheng-En Wu, Chien-Hung Chen, Yi-Ming Chan, and Chu-Song Chen. Compacting, picking and growing for unforgetting continual learning. *Advances in Neural Information Processing Systems*, 32, 2019. 2

[17] Hossein Jandaghi, Ali Divsalar, and Saeed Emami. The categorized orienteering problem with count-dependent profits. *Applied Soft Computing*, 113:107962, 2021. 1

[18] Yoonwoo Jeong, Seokjun Ahn, Christopher Choy, Anima Anandkumar, Minsu Cho, and Jaesik Park. Self-calibrating neural radiance fields. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5846–5854, 2021. 3, 5

[19] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42 (4), 2023. 1, 2

[20] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the national academy of sciences*, 114(13):3521–3526, 2017. 5

[21] Muhammed Kocabas, Jen-Hao Rick Chang, James Gabriel, Oncel Tuzel, and Anurag Ranjan. Hugs: Human gaussian splats. *arXiv preprint arXiv:2311.17910*, 2023. 2

[22] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural networks. *Communications of the ACM*, 60(6):84–90, 2017. 6

[23] kwea123. ngp-pl: a pytorch implementation of instant-ngp, 2022. 2

[24] Zhe Li, Zerong Zheng, Lizhen Wang, and Yebin Liu. Animatable gaussians: Learning pose-dependent gaussian maps for high-fidelity human avatar modeling. *arXiv preprint arXiv:2311.16096*, 2023. 2

[25] Chen-Hsuan Lin, Wei-Chiu Ma, Antonio Torralba, and Simon Lucey. Barf: Bundle-adjusting neural radiance fields. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5741–5751, 2021. 3

[26] David Lopez-Paz and Marc'Aurelio Ranzato. Gradient episodic memory for continual learning. *Advances in neural information processing systems*, 30, 2017. 2

[27] Jonathon Luiten, Georgios Kopanas, Bastian Leibe, and Deva Ramanan. Dynamic 3d gaussians: Tracking by persistent dynamic view synthesis. *arXiv preprint arXiv:2308.09713*, 2023. 2

[28] Arun Mallya and Svetlana Lazebnik. Packnet: Adding multiple tasks to a single network by iterative pruning. In *Proceedings of the IEEE conference on Computer Vision and Pattern Recognition*, pages 7765–7773, 2018. 2

[29] Arun Mallya, Dillon Davis, and Svetlana Lazebnik. Piggyback: Adapting a single network to multiple tasks by learning to mask weights. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 67–82, 2018. 2

[30] Ricardo Martin-Brualla, Noha Radwan, Mehdi SM Sajjadi, Jonathan T Barron, Alexey Dosovitskiy, and Daniel Duckworth. Nerf in the wild: Neural radiance fields for unconstrained photo collections. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7210–7219, 2021. 2

[31] Leandro do C Martins, Rafael D Tordecilla, Juliana Castaneda, Angel A Juan, and Javier Faulin. Electric vehicle routing, arc routing, and team orienteering problems in sustainable transportation. *Energies*, 14(16):5131, 2021. 1

[32] Ben Mildenhall, Pratul P Srinivasan, Matthew Tancik, Jonathan T Barron, Ravi Ramamoorthi, and Ren Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. *Communications of the ACM*, 65(1):99–106, 2021. 1, 2, 3, 5

[33] Ben Mildenhall, Peter Hedman, Ricardo Martin-Brualla, Pratul P Srinivasan, and Jonathan T Barron. Nerf in the dark: High dynamic range view synthesis from noisy raw images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16190–16199, 2022. 2

[34] Thomas Müller, Alex Evans, Christoph Schied, and Alexander Keller. Instant neural graphics primitives with a multiresolution hash encoding. *ACM Transactions on Graphics (ToG)*, 41(4):1–15, 2022. 2, 3

[35] Michael Niemeyer and Andreas Geiger. Giraffe: Representing scenes as compositional generative neural feature fields. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11453–11464, 2021. 2

[36] Michael Oechsle, Songyou Peng, and Andreas Geiger. Unisurf: Unifying neural implicit surfaces and radiance fields for multi-view reconstruction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5589–5599, 2021. 2

[37] Keunhong Park, Utkarsh Sinha, Jonathan T Barron, Sofien Bouaziz, Dan B Goldman, Steven M Seitz, and Ricardo Martin-Brualla. Nerfies: Deformable neural radiance fields. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5865–5874, 2021. 2

[38] Robert Pěnička, Jan Faigl, Petr Váňa, and Martin Saska. Dubins orienteering problem. *IEEE Robotics and Automation Letters*, 2(2):1210–1217, 2017. 1

[39] Ryan Po, Zhengyang Dong, Alexander W Bergman, and Gordon Wetzstein. Instant continual learning of neural radiance fields. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 3334–3344, 2023. 1, 2, 3, 4, 5

[40] Albert Pumarola, Enric Corona, Gerard Pons-Moll, and Francesc Moreno-Noguer. D-nerf: Neural radiance fields for dynamic scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10318–10327, 2021. 2

[41] Daniel Rebain, Wei Jiang, Soroosh Yazdani, Ke Li, Kwang Moo Yi, and Andrea Tagliasacchi. Derf: Decomposed radiance fields. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14153–14161, 2021. 2

[42] Barbara Roessle, Jonathan T Barron, Ben Mildenhall, Pratul P Srinivasan, and Matthias Nießner. Dense depth priors for neural radiance fields from sparse input views. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12892–12901, 2022. 2

[43] David Rolnick, Arun Ahuja, Jonathan Schwarz, Timothy Lillicrap, and Gregory Wayne. Experience replay for continual learning. *Advances in Neural Information Processing Systems*, 32, 2019. 2

[44] Antoni Rosinol, John J Leonard, and Luca Carlone. Nerf-slam: Real-time dense monocular slam with neural radiance fields. *arXiv preprint arXiv:2210.13641*, 2022. 5

[45] Johannes L Schonberger and Jan-Michael Frahm. Structure-from-motion revisited. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4104–4113, 2016. 3

[46] Hanul Shin, Jung Kwon Lee, Jaehong Kim, and Jiwon Kim. Continual learning with deep generative replay. *Advances in neural information processing systems*, 30, 2017. 2

[47] Matthew Tancik, Vincent Casser, Xinchen Yan, Sabeek Pradhan, Ben Mildenhall, Pratul Srinivasan, Jonathan T. Barron, and Henrik Kretzschmar. Block-NeRF: Scalable large scene neural view synthesis. *arXiv*, 2022. 5, 7

[48] Jiaxiang Tang, Jiawei Ren, Hang Zhou, Ziwei Liu, and Gang Zeng. Dreamgaussian: Generative gaussian splatting for efficient 3d content creation. *arXiv preprint arXiv:2309.16653*, 2023. 2

[49] Pieter Vansteenwegen and Aldy Gunawan. Orienteering problems. *EURO Advanced Tutorials on Operational Research*, 2019. 1

[50] Pieter Vansteenwegen, Wouter Souffriau, and Dirk Van Oudheusden. The orienteering problem: A survey. *European Journal of Operational Research*, 209(1):1–10, 2011. 1

[51] Peng Wang, Lingjie Liu, Yuan Liu, Christian Theobalt, Taku Komura, and Wenping Wang. Neus: Learning neural implicit surfaces by volume rendering for multi-view reconstruction. *arXiv preprint arXiv:2106.10689*, 2021. 2

[52] Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE transactions on image processing*, 13(4):600–612, 2004. 6

[53] Zirui Wang, Shangzhe Wu, Weidi Xie, Min Chen, and Victor Adrian Prisacariu. Nerf–: Neural radiance fields without known camera parameters. *arXiv preprint arXiv:2102.07064*, 2021. 3, 5

[54] Yi Wei, Shaohui Liu, Yongming Rao, Wang Zhao, Jiwen Lu, and Jie Zhou. Nerfingmvs: Guided optimization of neural radiance fields for indoor multi-view stereo. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5610–5619, 2021. 2

[55] Fanbo Xiang, Zexiang Xu, Milos Hasan, Yannick Hold-Geoffroy, Kalyan Sunkavalli, and Hao Su. Neutex: Neural texture mapping for volumetric neural rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7119–7128, 2021. 2

[56] Ju Xu and Zhanxing Zhu. Reinforced continual learning. *Advances in Neural Information Processing Systems*, 31, 2018. 2

[57] Ziyi Yang, Xinyu Gao, Wen Zhou, Shaohui Jiao, Yuqing Zhang, and Xiaogang Jin. Deformable 3d gaussians for high-fidelity monocular dynamic scene reconstruction. *arXiv preprint arXiv:2309.13101*, 2023. 2

[58] Lior Yariv, Jiatao Gu, Yoni Kasten, and Yaron Lipman. Volume rendering of neural implicit surfaces. *Advances in Neural Information Processing Systems*, 34:4805–4815, 2021. 2

[59] Lin Yen-Chen, Pete Florence, Jonathan T Barron, Alberto Rodriguez, Phillip Isola, and Tsung-Yi Lin. inerf: Inverting neural radiance fields for pose estimation. In *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 1323–1330. IEEE, 2021. 3

[60] Linfeng Zhang, Jiebo Song, Anni Gao, Jingwei Chen, Chenglong Bao, and Kaisheng Ma. Be your own teacher: Improve the performance of convolutional neural networks via self distillation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 3713–3722, 2019. 2, 3

[61] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 586–595, 2018. 6